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l₀ $x=7, y=5$ $\{ \alpha$

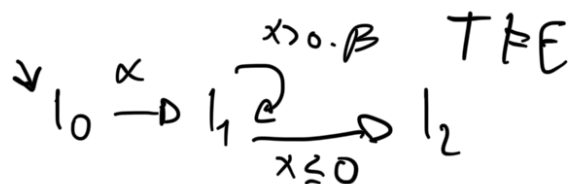
l₁ while $x > 0$ do

$x = x - 1$ $\} \beta$

od $y = y + 1$ $\}$

l₂ ...

never l₂ $\wedge$ y is odd $\models E$



$\langle l_0, \dots \rangle \rightarrow \langle l_1, x=7, y=5 \rangle$

$\downarrow$
 $\langle l_1, x=6, y=6 \rangle$

$\downarrow$
 $\dots$
 $\downarrow$

$\langle l_1, x=0, y=12 \rangle$

$\downarrow$
 $\langle l_2, x=0, y=12 \rangle$

$f: S \rightarrow \hat{S}$

$P(s) \Rightarrow f(s')$ if $s \rightarrow s'$

l_0, l_1, l_2

$x = 0$

$x \equiv_2 y$ $x-y$ is even

$x \equiv_2 y \wedge x=0 \Rightarrow y$ even

$\langle l_0, \dots \rangle$ $T' \models E$

$\downarrow$
 $\langle l_1, x \neq 0 \rangle$

$x \equiv_2 y$

$\downarrow$

$\langle l_1, x=0 \rangle \rightarrow \langle l_2, x=0 \rangle$
 $x \equiv_2 y$ $x \equiv_2 y$

LT-properties:

- safety - mutual exclusion, deadlock freedom

- liveness

$\Phi \models \text{true} \wedge \Phi_1 \wedge \Phi_2 \mid \neg \Phi \mid \Phi_1 \vee \Phi_2 \mid \Phi_1 \Rightarrow \Phi_2 \mid \dots$

$\varphi \dots \in AP$

$$A \subseteq AP$$

$$\{a, b\} \not\models (a \Rightarrow \neg b) \vee c$$

$$A \models \text{tt}$$

$$\{a, b\} \models a \vee c$$

$$A \models a \Leftrightarrow a \in A$$

$$A \models \phi_1 \wedge \phi_2 \Leftrightarrow A \models \phi_1 \wedge A \models \phi_2$$

$$A \models \neg \phi \Leftrightarrow A \not\models \phi$$

For states s of a TS over AP : $s \models \phi \Leftrightarrow L(s) \models \phi$

Let E be an LT property over AP .

E is called an **invariant** if there exists a propositional formula ϕ over AP such that

$$E = \{A_0 A_1 A_2 \dots \in (2^{AP})^\omega : \forall i \geq 0. A_i \models \phi\}$$

ϕ is called the **invariant condition** of E .

$M \cup T \in X =$ set of all infinite words $A_0 A_1 A_2 \dots$ s.t.
 $\forall i \in \mathbb{N}. \text{crit}_1 \notin A_i \text{ or } \text{crit}_2 \notin A_i$

$$A = \{\text{crit}_1, \text{crit}_2, \dots\} \quad \phi = \neg \text{crit}_1 \vee \neg \text{crit}_2$$

Let E be an LT property over AP . E is called an invariant if there exists a propositional formula Φ s.t.

$$E = \{ A_0 A_1 A_2 \dots \in (2^{AP})^\omega : \forall i \geq 0. A_i \models \Phi \}$$

Let $\mathcal{T}$ be a TS over AP without terminal states. Then:

$$\begin{aligned} \mathcal{T} \models E & \text{ iff } \text{trace}(\pi) \in E \text{ for all } \pi \in \text{Paths}(\mathcal{T}) \\ & \text{ iff } s \models \Phi \text{ for all states } s \text{ on a path of } \mathcal{T} \\ & \text{ iff } s \models \Phi \text{ for all states } s \in \text{Reach}(\mathcal{T}) \end{aligned}$$

↳ {reachable states}

i.e., Φ holds in all initial states and is **invariant** under all transitions

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DFS-based invariant checking

LTPROP/IS2.5-7

input: finite transition system $\mathcal{T}$, invariant condition Φ

$U := \emptyset \leftarrow$ stores the “processed” states

$\pi := \emptyset \leftarrow$ stack for error indication

FOR ALL $s_0 \in S_0$ DO

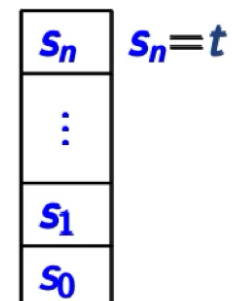
IF $DFS(s_0, \Phi)$ THEN

return “no” and $reverse(\pi)$

FI

OD

return “yes”



$DFS(s_0, \Phi)$ returns “true” iff depth-first search from state s_0 leads to some state t with $t \not\models \Phi$

Definition of safety properties

IS2.5-11

Let E be a LT property over AP , i.e., $E \subseteq (2^{AP})^\omega$.

E is called a safety property if for all words

$$\sigma = A_0 A_1 A_2 \dots \in (2^{AP})^\omega \setminus E$$

there exists a finite prefix $A_0 A_1 \dots A_n$ of σ such that *none* of the words $A_0 A_1 \dots A_n B_{n+1} B_{n+2} B_{n+3} \dots$ belongs to E , i.e.,

$$E \cap \{\sigma' \in (2^{AP})^\omega : A_0 \dots A_n \text{ is a prefix of } \sigma'\} = \emptyset$$

Such words $A_0 A_1 \dots A_n$ are called **bad prefixes** for E .

E = set of all infinite words that
do *not* have a **bad prefix**

$$\text{BadPref}_E = \{ \text{bad prefixes for } E \} \subseteq (2^{AP})^+$$

minimal bad prefixes = any word $A_0 \dots A_i \dots A_n \in \text{BadPref}$
s.t. no proper prefix $A_0 \dots A_i$ is a bad
prefix for E

Satisfaction of safety properties

IS2.5-11A

Let $E \subseteq (2^{AP})^\omega$ be a safety property, T a TS over AP .

$$T \models E \text{ iff } \text{Traces}(T) \subseteq E$$

$$\text{iff } \text{Traces}_{fin}(T) \cap \text{BadPref} = \emptyset$$

$$\text{iff } \text{Traces}_{fin}(T) \cap \text{MinBadPref} = \emptyset$$

BadPref = set of all **bad prefixes** of E

MinBadPref = set of all minimal bad prefixes of E
 $\text{Traces}(\mathcal{T})$ = set of traces of $\mathcal{T}$
 $\text{Traces}_{\text{fin}}(\mathcal{T})$ = set of finite traces of $\mathcal{T}$
 $= \{ \text{trace}(\hat{\pi}) : \hat{\pi} \text{ is an initial, finite path fragment of } \mathcal{T} \}$

$\forall$ invariant is safety property

$\emptyset$ is a safety property

$(2^{AP})^w$ is a safety property

$$\text{pref}(\sigma) = \{ A_0 A_1 \dots A_n : n \geq 0 \}$$

$\downarrow$
 infinite
 word $A_0 \dots$

$$\text{for } E \subseteq (2^{AP})^w, \text{pref}(E) = \bigcup_{\sigma \in E} \text{pref}(\sigma)$$

$$\text{cl}(E) = \{ \sigma \in (2^{AP})^w : \text{pref}(\sigma) \subseteq \text{pref}(E) \}$$

$$E \text{ is safety property} \iff \text{cl}(E) = E$$

Safety and finite trace inclusion

IS2.5-SAFETY-TRACEFIN

remind: LT properties and trace inclusion:

If $\mathcal{T}_1$ and $\mathcal{T}_2$ are TS over AP then:

$$\text{Traces}(\mathcal{T}_1) \subseteq \text{Traces}(\mathcal{T}_2)$$

iff for all LT properties E : $\mathcal{T}_2 \models E \implies \mathcal{T}_1 \models E$

safety properties and finite trace inclusion:

If $\mathcal{T}_1$ and $\mathcal{T}_2$ are TS over AP then:

$$\text{Traces}_{fin}(\mathcal{T}_1) \subseteq \text{Traces}_{fin}(\mathcal{T}_2)$$

iff for all safety properties E : $\mathcal{T}_2 \models E \implies \mathcal{T}_1 \models E$

safety properties and finite trace equivalence:

If $\mathcal{T}_1$ and $\mathcal{T}_2$ are TS over AP then:

$$\text{Traces}_{fin}(\mathcal{T}_1) = \text{Traces}_{fin}(\mathcal{T}_2)$$

iff $\mathcal{T}_1$ and $\mathcal{T}_2$ satisfy the same safety properties

Suppose that $\mathcal{T}$ and $\mathcal{T}'$ are TS over AP such that

(1) $\mathcal{T}$ has no terminal states

(2) $\mathcal{T}'$ is finite

image-finiteness
is sufficient

(3) $\text{Traces}_{fin}(\mathcal{T}) \subseteq \text{Traces}_{fin}(\mathcal{T}')$

Then $\text{Traces}(\mathcal{T}) \subseteq \text{Traces}(\mathcal{T}')$

$$\sigma = A_0 A_1 A_2 \dots \in \text{Traces}(\mathcal{T})$$

$$\sigma^n = A_0 A_1 A_2 \dots A_n \in \text{Trace}_{fin}(\mathcal{T}) \Rightarrow \sigma^n \in \text{Trace}_{fin}(\mathcal{T}')$$

$$\pi = s_0 s_1 \dots \in \text{Paths}(\mathcal{T})$$

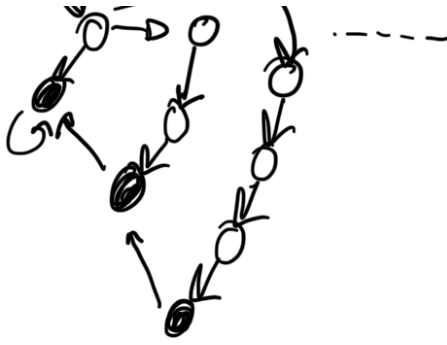
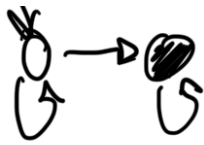
$$\text{Trace}(\pi) = \sigma$$

$$\Rightarrow \hat{\pi}^n \in \text{Paths}_{fin}(\mathcal{T}')$$

$$\pi^n \text{ s.t., } \text{trace}(\pi^n) = \sigma^n$$

$$\text{trace}(\hat{\pi}^n) = \sigma^n$$





$$\hat{\pi}^0 = S_0$$

$$\hat{\pi}_1 = S_0^1 S_1^1$$

$$\hat{\pi}_2 = S_0^2 S_1^2 S_2^2$$